# Phishing Email Detection & Awareness System


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**CONFIDENTIAL SECURITY ASSESSMENT**

# Executive Summary

This report presents the analysis of phishing email samples to identify common attack patterns and improve user awareness of email-based threats. Phishing attacks are one of the most common cybersecurity risks, targeting users through deceptive messages designed to steal sensitive information.

Four phishing email samples were analyzed during this assessment. Each email was examined based on sender identity, message content, and embedded links to identify potential red flags and malicious indicators.

All analyzed emails were classified as phishing due to the presence of multiple indicators such as suspicious links, generic greetings, and urgency-based language. These techniques are commonly used by attackers to manipulate users into taking harmful actions.

The goal of this report is not only to identify phishing attempts but also to provide clear and simple guidance that helps users recognize and avoid such attacks in real-world scenarios.

In addition to identifying phishing indicators, this assessment highlights how attackers exploit human behavior rather than technical weaknesses. By using techniques such as urgency, impersonation, and deceptive links, attackers increase the likelihood of user interaction.

This report emphasizes the importance of user awareness as a critical layer of defense. Educating users to recognize phishing attempts can significantly reduce the success rate of such attacks and strengthen the overall security posture of an organization.

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# Email Analysis

This section analyzes phishing email samples to identify common attack patterns, classify risk levels, and explain how these attacks attempt to deceive users.

# Email 1: Bank Account Alert

**From:** support@secure-bank-login.com
**To:** user@example.com
**Subject:** Urgent: Your Account Will Be Locked
Email Body:
Dear Customer,
We detected unusual activity on your bank account. To prevent temporary suspension, please verify your identity immediately.
**Verify Now: http://secure-bank-login[.]com**
Failure to act within 24 hours will result in account suspension.
Regards,
Security Team

| Field | Analysis |
|---|---|
| **Classification** | Phishing |
| **Sender Analysis** | The sender uses a suspicious domain that does not belong to a legitimate bank. |
| **Message Analysis** | The message creates urgency and fear to pressure the user into taking immediate action. |
| **Link Analysis** | The link is not a real bank domain and is likely malicious. |
| **Phishing Indicators** | Generic greeting, urgency, suspicious link, request for sensitive information |
| **Explanation** | This email attempts to trick users into entering personal or banking information on a fake website. |

# Email 2: Microsoft Security Alert

**From:** security@microsoft-secure-login.net
**To:** user@example.com
**Subject:** Microsoft Security Alert
**Email Body:**
Hello,
Your Microsoft account password is set to expire today. Please confirm your account details to avoid losing access.

**Update Password: http://microsoft-secure-login[.]net**

Thank you,
Microsoft Support

| Field | Analysis |
|---|---|
| **Classification** | Phishing |
| **Sender Analysis** | The sender impersonates Microsoft using a fake domain. |
| **Message Analysis** | The email creates urgency by stating the password expires immediately. |
| **Link Analysis** | The link is not an official Microsoft domain and is likely malicious. |
| **Phishing Indicators** | Generic greeting, urgency, threat of losing access, suspicious link |
| **Explanation** | This email tricks users into entering login credentials on a fake Microsoft page. |

# Email 3: Delivery Failure Notice

**From:** support@delivery-confirmation.xyz
**To:** user@example.com
**Subject:** Delivery Failed – Action Required
**Email Body:**
Dear User,
We attempted to deliver your package but failed. Please confirm your address to reschedule delivery.
**Track Package: http://delivery-confirmation[.]xyz**
Thank you,
Delivery Service

| Field | Analysis |
|---|---|
| **Classification** | Phishing |
| **Sender Analysis** | The sender does not belong to a known delivery company. |
| **Message Analysis** | The message is vague and lacks important delivery details. |
| **Link Analysis** | The domain uses .xyz which is uncommon and suspicious. |
| **Phishing Indicators** | Generic greeting, suspicious domain, lack of details, request for personal information |
| **Explanation** | This email attempts to steal user information by pretending to be a delivery service. |

# Email 4: Job Offer Attachment Scam

**From:** hr@career-opportunities-global.com
**To:** user@example.com
**Subject:** Job Opportunity – Immediate Hiring
**Email Body:**
Dear Applicant,
We are pleased to inform you that you have been shortlisted for a job opportunity with our company.
Please review the attached document for job details and salary information.
**Download Attachment: Job_Offer_Details.zip**
Kindly respond within 24 hours to confirm your interest.
Best regards,
HR Department

| Field | Analysis |
|---|---|
| **Classification** | Phishing |
| **Sender Analysis** | The sender uses a generic domain that does not belong to a known company. |
| **Message Analysis** | The email creates excitement and urgency to pressure the user into opening the attachment. |
| **Link Analysis** | The attachment is a compressed file (.zip), which may contain malicious software. |
| **Phishing Indicators** | Unexpected job offer, urgency, unknown sender, suspicious attachment |
| **Explanation** | This email attempts to trick users into downloading a malicious file that could install malware on their system. |

# Observations

The analysis of the phishing email samples reveals consistent attack patterns used by cyber attackers. Most emails rely on social engineering techniques such as urgency, fear, or excitement to manipulate user behavior.

A common trend across all emails is the use of generic greetings and lack of personalization, which indicates mass-targeted attacks rather than legitimate communication. Additionally, all emails contain either suspicious links or malicious attachments designed to steal sensitive information or compromise user systems.

Another key observation is the use of fake or misleading domains that attempt to imitate trusted organizations. These domains are crafted to appear legitimate at first glance but differ from official domains upon closer inspection.
Overall, the emails demonstrate that phishing attacks are designed to exploit human trust rather than technical vulnerabilities, making user awareness a critical factor in preventing such threats.

# Awareness & Prevention

## Do's

- Check sender email carefully
- Hover over links before clicking
- Verify requests through official websites
- Report suspicious emails

## Don'ts

- Do not click unknown links
- Do not share passwords or OTPs
- Do not trust urgent messages
- Do not download unexpected attachments

# Conclusion

This report analyzed multiple phishing email samples and identified common techniques used by attackers, including urgency, impersonation, and deceptive links. All analyzed emails were classified as phishing due to multiple strong indicators.

Improving user awareness is essential to reducing phishing risks. By recognizing suspicious patterns and following basic security practices, users can prevent most phishing attacks.

In addition, this assessment highlights that phishing attacks continue to evolve by using different approaches such as malicious links and harmful attachments. Attackers adapt their techniques to target users in various scenarios, including banking, account security, delivery services, and job opportunities.

Organizations must prioritize continuous user awareness and training programs to reduce the risk of successful phishing attacks. Regular testing, simulated phishing campaigns, and clear reporting procedures can significantly improve an organization’s ability to detect and respond to threats.

By combining user education with proper security practices, organizations can build a stronger defense against phishing attacks and protect sensitive information from unauthorized access.